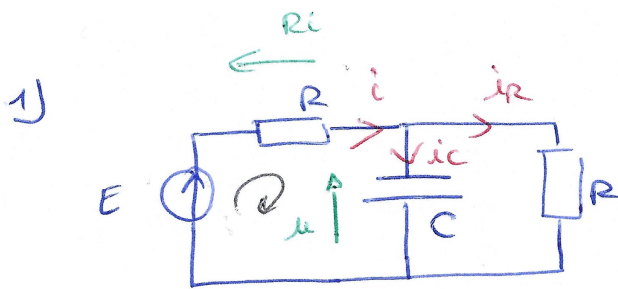


Etude de circuits électriques



$$i = i_c + i_R$$

On a (la 1^{re} maille) :

$$E - Ri - u = 0$$

$$i_c = C \frac{du}{dt}$$

$$E - R \left(\frac{u}{R} + C \frac{du}{dt} \right) - u = 0$$

$$i_R = \frac{u}{R}$$

$$E - 2u - RC \frac{du}{dt} = 0$$

$$i = i_c + i_R$$

$$RC \frac{du}{dt} + 2u = E$$

$$= \frac{u}{R} + C \frac{du}{dt}$$

$$\frac{du}{dt} + \frac{2u}{RC} = \frac{E}{RC}$$

2) On pose $\tau = \frac{RC}{2}$

$$\frac{du}{dt} + \frac{u}{\tau} = \frac{E}{2\tau}$$

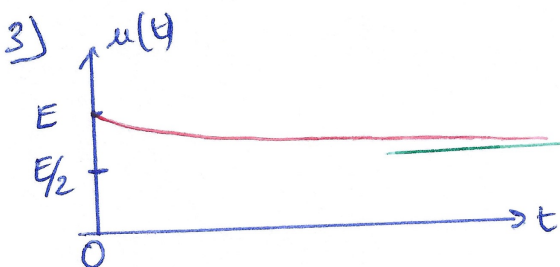
$$u(t) = u_h(t) + u_p(t) \\ = Ce^{-t/\tau} + \frac{E}{2}$$

2 $\frac{u_p}{\tau} = \frac{E}{2\tau}$

$$u_p = \frac{E}{2}$$

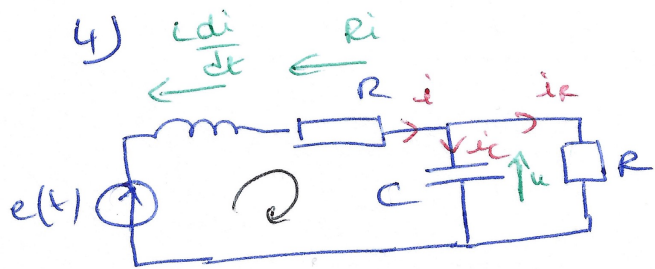
à $t = 0$ $u(0) = E \Rightarrow C + \frac{E}{2} = E \Rightarrow C = \frac{E}{2}$

$$u(t) = \frac{E}{2} (1 + e^{-t/\tau})$$



par $t \rightarrow +\infty$
 $u(t) \rightarrow \frac{E}{2}$

C'est 1 exp. dec.



Loi des mailles: $e(t) - L \frac{di}{dt} - Ri - u = 0$

$$i_C = C \frac{du}{dt}$$

$$i_R = \frac{u}{R}$$

$$i = i_C + i_R$$

$$e(t) - L \frac{d}{dt} \left(\frac{u}{R} + C \frac{du}{dt} \right) - R \left(\frac{u}{R} + C \frac{du}{dt} \right) - u = 0$$

$$e(t) = \frac{L}{R} \frac{du}{dt} + LC \frac{d^2 u}{dt^2} + u + RC \frac{du}{dt} + u$$

$$e(t) = LC \frac{d^2 u}{dt^2} + \left(\frac{L}{R} + RC \right) \frac{du}{dt} + 2u$$

Forme
canonique

$$\frac{d^2 u}{dt^2} + \left(\frac{1}{RC} + \frac{R}{L} \right) \frac{du}{dt} + 2 \frac{u}{LC} = \frac{e(t)}{LC}$$

5] On a $\omega_0^2 = \frac{2}{LC} \Rightarrow \omega_0 = \sqrt{\frac{2}{LC}}$

$$\frac{\omega_0}{\Phi} = \frac{1}{RC} + \frac{R}{L} \Rightarrow \Phi = \frac{\omega_0}{\frac{1}{RC} + \frac{R}{L}} = \frac{\sqrt{2}}{\sqrt{LC} \left(\frac{1}{RC} + \frac{R}{L} \right)}$$

$$= \frac{\sqrt{2LC}}{LC \left(\frac{1}{RC} + \frac{R}{L} \right)}$$

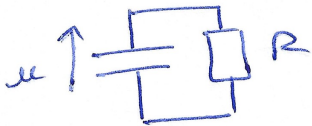
$$\Phi = \frac{\sqrt{2LC}}{\frac{L}{R} + RC}$$

6) A $t=0^-$ $u(0^-)=0$
 $i(0^-)=i_c(0^-)=i_R(0^-)=0$

donc à $t=0^+$: $u(0^+)=0$ par cte de u dans C .
 $i(0^+)=0$ " i dans L

à $t=0^+$

$u(0^+)=0 \Rightarrow R i_R(0^+)=0 \Rightarrow i_R(0^+)=0$



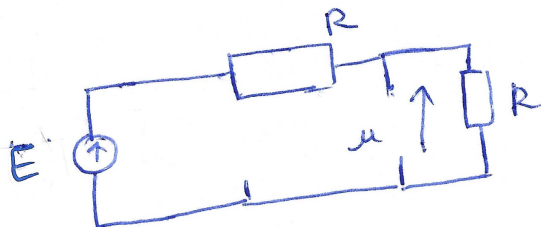
$\Rightarrow i_c(0^+)=i(0^+)-i_R(0^+)=0$

$\Rightarrow \frac{du(0^+)}{dt}=0$ car $i_c = C \frac{du}{dt}$

En régime permanent :

1/3 $i_c = C \frac{du}{dt} = 0$

$u_L = L \frac{di}{dt} = 0$



Port diviseur de tension.

$u = \frac{E}{2}$

7) $Q > 1/2$ réponse pseudo-périodique

2) $u(t) = (A \cos \Omega t + B \sin \Omega t) e^{-\frac{\omega_0}{2Q} t} + \frac{E}{2}$

$\Omega = \omega_0 \sqrt{1 - \frac{1}{4Q^2}}$

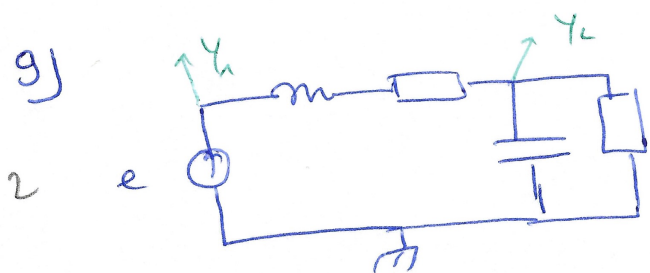
8) Avec les CI $u(0^+)=0 = \frac{du(0^+)}{dt}$

2 On a $A = -\frac{E}{2}$
 $B = -\omega_0 (4Q\Omega) \times E$

2

$$u(t) = \frac{E}{2} \left(1 - e^{-\frac{\omega_0 t}{2Q}} \left(\frac{\omega_0}{2Q\Omega} \sin(\Omega t) + \cos \Omega t \right) \right)$$

9)



10) $u_\infty = \frac{E}{2} = 10V \Rightarrow \underline{E = 20V}$

11) $Q \approx 10$ (to oscillate with).

B.

12) $e = E_0 e^{j\omega t}$

$u = u_m e^{j\omega t}$

$u_m = u_m e^{j\varphi}$

13) (a) $\frac{d^2 u}{dt^2} + \frac{\omega_0}{Q} \frac{du}{dt} + \omega_0^2 u = e(t) \frac{\omega_0^2}{2}$

$\frac{d}{dt} \leadsto \times j\omega \Rightarrow -\omega^2 u + \frac{\omega_0}{Q} j\omega u + \omega_0^2 u = \underline{e} \cdot \frac{\omega_0^2}{2}$

$\frac{d^2}{dt^2} \leadsto \times -\omega^2$

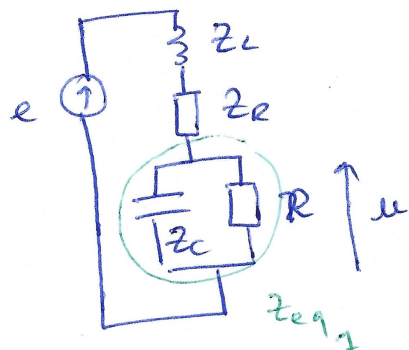
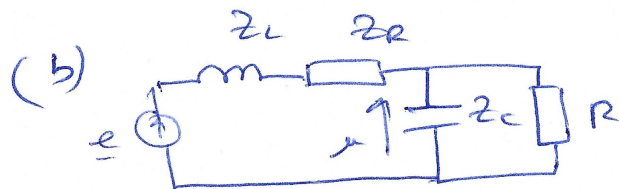
$(\omega_0^2 - \omega^2 + \frac{\omega_0}{Q} j\omega) u = \underline{e} \omega_0^2 / 2$

$u = \frac{\underline{e} \omega_0^2 / 2}{\omega_0^2 - \omega^2 + j \frac{\omega \omega_0}{Q}}$

(donc $|u_m| = u_m = \frac{|\underline{e}|}{\sqrt{(\omega_0^2 - \omega^2)^2 + \left(\frac{\omega \omega_0}{Q}\right)^2}} u = \frac{e/2}{1 + j \frac{\omega}{Q \omega_0} - \frac{\omega^2}{\omega_0^2}}$

$= \frac{E_m}{\sqrt{(\omega_0^2 - \omega^2)^2 + \left(\frac{\omega \omega_0}{Q}\right)^2}} = \frac{e/2}{1 + j \frac{\omega}{Q} - \omega^2}$

$$\underline{u}_m = \frac{1/2}{1 + x^2 + j \frac{x}{Q}} \underline{E}_0$$



$$\frac{1}{z_{eq1}} = \frac{1}{R} + jC\omega$$

Pont diviseur de tension.

$$\underline{u} = \frac{z_{eq1}}{z_{eq1} + z_R + z_L} \underline{e}$$

$$\underline{u} = \frac{1}{1 + \frac{1}{z_{eq1}} (z_R + z_L)} \underline{e}$$

$$= \frac{1}{1 + \left(\frac{1}{R} + jC\omega \right) (R + jL\omega)} \underline{e}$$

$$= \frac{1}{1 + 1 + j\frac{L\omega}{R} + jRC\omega - LC\omega^2} \underline{e}$$

$$= \frac{1}{2 + j\left(\frac{L}{R} + RC \right) \omega - LC\omega^2} \underline{e}$$

$$= \frac{1/2}{1 + j\left(\frac{L}{R} + RC \right) \frac{\omega}{2} - \left(\frac{LC}{2} \right) \omega^2} \underline{e}$$

$= 1/\varphi\omega_0$

$$\underline{u} = \frac{1/2}{1 + j\frac{\omega}{\phi\omega_0} - \left(\frac{\omega}{\omega_0}\right)^2} \underline{e}$$

$$\underline{u_M} = \frac{1/2}{1 - u^2 + j\frac{u}{\phi}} E_0$$

14.

$$u_M = |\underline{u_M}| = \frac{E_0/2}{\sqrt{(1-u^2)^2 + \left(\frac{u}{\phi}\right)^2}}$$

$$\begin{aligned} \varphi = \arg(\underline{u_M}) &= \arg(E_0/2) - \arg\left(1 - u^2 + j\frac{u}{\phi}\right) \\ &= -\arg\left(1 - u^2 + j\frac{u}{\phi}\right). \end{aligned}$$

si $1 - u^2 > 0$
ie $u < 1$

$$\varphi = -\text{Arctan} \frac{u/\phi}{1 - u^2}$$

si $1 - u^2 < 0$
ie $u > 1$

$$\varphi = -\text{Arctan} \frac{u/\phi}{1 - u^2} - \pi$$

si $u = 1$

$$\varphi = -\pi/2$$

15. u_M atteint 1 maximum pour ω proche de ω_0 .

C'est le phénomène de résonance en tension.

$\left\{ \begin{array}{l} \omega_0 = \omega_{\text{res}} \\ \text{pic } \varphi \end{array} \right.$

$\Rightarrow$ résonance agée de Q élevée.

16. BF $\varphi \xrightarrow{u \rightarrow 0} -\text{Arctan } 0 = 0$

$$u = 1$$

$$\varphi = -\pi/2$$

HF $\varphi \xrightarrow{u \rightarrow +\infty} -\pi = -180^\circ$

courbe conforme aux prédictions

217. A BF $u_M \xrightarrow{u \rightarrow 0} E_0/2 = 1V \Rightarrow E_0 = 2V$